

Solinteg EMS BattCtrl

Reverse engineered by @sixuniform

Note: This document describes the observed behavior of **EMS BattCtrl** on Solinteg **MHT firmware 9.x**. Some behaviors differ from the official Modbus documentation.

Overview

EMS BattCtrl allows an external Energy Management System (EMS) to control the inverter's AC power exchange while allowing the inverter to continue managing:

- Battery safety (BMS limits)
- SOC limits
- Charge/discharge current limits
- PV MPPT
- Grid synchronization

The EMS provides a target and operating limits, while the inverter determines how PV and battery power are combined.

Registers

Register	Name	Observed Function
50207	EMS BattCtrl Charge/Discharge Target	Desired inverter AC power (signed)
50208	EMS BattCtrl Max Grid Export	Maximum positive inverter AC power
50209	EMS BattCtrl Max Grid Import	Maximum negative inverter AC power

Important

Registers **50208** and **50209** must be written **after** the inverter has entered EMS BattCtrl mode. Writing them before switching to BattCtrl appears to have no effect.

It also seems these constraint settings is available to use for EMS General mode, despite the Modbus documentation specifying them only for EMS BattCtrl.

Modbus Register Format

All EMS BattCtrl power registers are **signed 16-bit integers (S16)** with a **scale factor of 0.01 kW/count**.

Engineering Value (kW) = Register Value × 0.01
Register Value = kW × 100

Examples:

Desired Value	Register Value
10.00 kW	1000
6.50 kW	650
2.00 kW	200
1.00 kW	100
0.00 kW	0
-1.00 kW	-100
-2.00 kW	-200
-10.00 kW	-1000

Registers:

Register	Type	Scale	Description
50207	S16	0.01 kW EMS BattCtrl Charge/Discharge Target	
50208	S16	0.01 kW EMS BattCtrl Maximum Inverter AC Export	
50209	S16	0.01 kW EMS BattCtrl Maximum Inverter AC Import	
50210	U16	Enum Output Priority (0 = PV, 1 = Battery)	

Examples:

Request 6 kW inverter output

50207 = 600

Limit inverter export to 2 kW

50208 = 200

Limit inverter import to 2 kW

50209 = -200

PV Priority

50210 = 0

Battery Priority

50210 = 1

Sign convention

Positive values:

- Inverter exports AC power.
- Battery is typically discharging.

Negative values:

- Inverter imports AC power.
- Battery is typically charging.

Example:

Request approximately **6 kW inverter output**.

`50207 = 600`

Request approximately **5 kW inverter input** (battery charging).

`50207 = -500`

What does 50207 control?

Despite its name ("Charge/Discharge Target"), firmware 9.x behaves as if **50207 controls inverter AC power**, not battery power.

The inverter then allocates this power between PV and battery according to the selected priority mode.

For example:

```
PV = 3 kW
Target = 6
kW PV
Priority
```

Result:

- PV = 3 kW
- Battery = 3 kW
- Inverter output = 6 kW

House loads do **not** affect the requested inverter output.

What does 50208 limit?

50208 limits the **maximum positive inverter AC power**.
It does **not** directly limit the site's net export.

Example:

PV = 3 kW
Battery capable of 6 kW discharge
50207 = 600
50208 = 200

PV Priority

Result:

- PV = 2 kW
- Battery = 0 kW
- Inverter output = 2 kW
- EMS CmdLim active
- PV PLim active

Battery Priority

Result:

- Battery = 2 kW
- PV = 0 kW
- Inverter output = 2 kW
- EMS CmdLim active
- PV PLim active

The inverter limits total AC output to 2 kW and allocates that power according to the selected priority.

What does 50209 limit?

50209 limits the **maximum inverter AC import**.

It does **not** include house loads.

Example:

Charge target = -700
PV = 3 kW
House load = 700 W
50209 = -200

Result:

- PV contributes 3 kW
- Grid contributes 2 kW
- Battery charges at 5 kW
- House imports an additional 700 W
- Utility meter shows approximately 2.7 kW import

The inverter respects its own 2 kW import limit while the house load is supplied independently.

Priority Modes

PV Priority

PV power is used first.

The battery supplies or absorbs the remaining power required to achieve the requested inverter power.

Example

Target = 10 kW
PV = 5 kW

Result:

- PV = 5 kW
 - Battery = 5 kW
 - Inverter output = 10 kW
-

Battery Priority

Battery power is used first.

PV may be curtailed if necessary.

Example

Target = 2 kW
PV available = 3 kW
Max Export = 2 kW

Result:

- Battery = 2 kW
 - PV = 0 kW
 - PV PLim active
-

Status indications

EMS CmdLim

The requested inverter power cannot be achieved because one or more EMS limits became active.

Typical causes:

- Max Grid Export reached
- Max Grid Import reached

PV PLim

PV production has been curtailed.

This commonly occurs when:

- Max Grid Export limits inverter output
 - Battery Priority is selected
 - PV production exceeds the permitted inverter output
 - The inverter is limiting PV for other reasons (like while Slow Loading after errors)
-

House loads

One important observation is that house loads are **not included** in the EMS import/export limits.

The limits apply only to the inverter's own AC power exchange.

Therefore:

- House consumption may increase grid import beyond the configured Max Grid Import.
- House consumption may reduce actual grid export below the configured inverter output.

The EMS should therefore consider house load when calculating the desired inverter power if a specific site import/export target is required.

Typical control sequence

1. Enable EMS BattCtrl mode.
2. Wait until the inverter reports BattCtrl active.
3. Write Max Grid Export (50208).
4. Write Max Grid Import (50209).
5. Write Charge/Discharge Target (50207).
6. Continue updating 50207 as required.

Writing 50208 or 50209 before entering BattCtrl may have no effect.

Practical examples

Zero inverter export

```
50208 = 0
50207 = 1000
```

Observed:

- PV curtailed
 - Battery idle
 - Inverter AC output = 0 W
 - House supplied from grid
-

Charge from PV only

```
50207 = -1000
50209 = 0
PV Priority
```

Example:

PV = 5 kW

Result:

- Battery charges at 5 kW
 - No grid power used for charging
 - House still imports its own load
-

Charge using PV + up to 2 kW grid

50207 = -1000
50209 = -200

Example:

PV = 3 kW

Result:

- PV contributes 3 kW
- Grid contributes up to 2 kW
- Battery charges at approximately 5 kW

Export limit of 2 kW

50207 = 1000
50208 = 200

PV Priority:

- PV used first
- Battery only if required
- Total inverter output limited to 2 kW

Battery Priority:

- Battery used first
- PV curtailed if necessary
- Total inverter output limited to 2 kW

Notes

The official Modbus manual refers to registers 50208 and 50209 as "Max Grid Export" and "Max Grid Import". On MHT firmware 9.x, observed behavior indicates these registers limit the inverter's own AC power exchange rather than the site's net import/export measured by the utility meter.

These observations are based on testing with:

- Solinteg MHT firmware 9.x
- EMS BattCtrl mode
- Native inverter metering
- Pylontech HV battery
- No EPS/Off-grid load connected